

PRE-SEED · \$1M · 2026

The Crypto Lending Platform That *Explains Itself.*

Crypto-collateralized lending infrastructure for credit unions — with
AI-powered underwriting guidance built in. Pre-Seed · \$1M · 2026

140M

CU MEMBERS IN THE US

\$2.2T

CU ASSETS UNDER MANAGEMENT

\$0

B2B CRYPTO LENDING INFRASTRUCTURE

SCROLL

01 · PROBLEM

Credit unions are *locked out* of crypto lending.

140 million Americans trust their credit union over any bank. But when those members want to unlock liquidity from their Bitcoin or Ethereum holdings, they're forced to go to D2C lenders like Nexo or Ledn — or sell, triggering a taxable event.

Credit unions want to serve these members. The regulatory complexity is the barrier — not the appetite.

- 01 No compliant crypto lending infrastructure exists for NCUA-regulated institutions
- 02 D2C competitors (Nexo, Ledn, SALT) aren't B2B — they compete with CUs, not serve them
- 03 Younger members with crypto assets are migrating to fintech — CUs are losing them
- 04 No risk assessment model purpose-built for crypto collateral under NCUA guidelines exists
- 05 Custody, bank connectivity, and KYC/AML integration requires months of compliance work per institution

02 · SOLUTION

The *full stack* a credit union needs to go live.

Every competitor is building faster pipes. Aetherum builds the explanation layer — the first crypto lending platform that tells a member why they can borrow what they can borrow, automatically, at the point of decision.

01 · Lending Engine

Crypto-Collateralized Loans

Members keep their BTC/ETH exposure, avoid the taxable sale, and receive USD liquidity at 8–12% APR. Real-time LTV monitoring with automated margin calls.

02 · Risk

DACS™ Risk Assessment

Dynamic Asset Credit Scoring — a patent-pending ten-pillar risk model purpose-built for crypto collateral. AI-powered guidance layer live — members receive plain-language borrowing insights directly from their DACS profile.

PATENT PENDING

03 · Custody

Institutional Custody

Each credit union holds its own direct institutional custody account. Member wallets live within the CU's structure. Aetherum is the software layer only — we never touch member assets.

04 · Compliance

Compliance Infrastructure

Bank-grade financial data connectivity approved and live across 6 core products. KYC/AML and fraud infrastructure MSA fully executed — production-ready compliance infrastructure live. Annual billing secured, consortium anonymization in contract.

05 · Compliance

ERC-3643 On-Chain Identity

Clean-room implementation of three ERC-3643 compliance modules — jurisdiction gating, collateral LTV verification, and CU member identity attestation. Built on the T-REX standard. Deployed on Sepolia testnet.

PATENT PENDING

06 · Privacy

Privacy-Preserving Underwriting (FHE-DACS)

Member financial data is encrypted before the risk assessment process begins. Lending decisions are computed without exposing raw inputs at any step — ever. Patent-pending architecture purpose-built for NCUA-regulated institutions.

PATENT PENDING

Staking Module

Members can stake supported digital assets directly through the platform — earning yield while maintaining collateral eligibility for crypto-backed loans. Adds a yield layer to the Ethereum product suite, deepening member engagement and creating an additional revenue stream for credit union partners. Coming Soon.

Tax-Position Score & Collateral Optimizer (TPSOE™)

The TPSOE engine scores the tax efficiency of every collateral position in a member's portfolio — ranking assets by embedded tax liability so the optimal pledge set is always selected. Members borrow smarter, not just more: no taxable event, no unnecessary capital gains exposure. The Collateral Optimizer computes UGLV and TAEV per asset, selects the lowest tax-cost collateral set that satisfies LTV requirements, and surfaces estimated tax savings to the member. Patent-pending sixth DACS pillar. No B2B competitor offers this to institutions.

PATENT PENDING

Member Crypto Insights

Real-time visibility into the digital assets your members already own — before they need a loan. Consent-based wallet connections, DACS eligibility scoring, and an opt-in loan pipeline that surfaces qualified borrowers before they apply. The first CU intelligence layer built for the digital asset era.

PATENT PENDING

Collateral Intelligence Engine (CIE™)

Every collateral position monitored 24/7. Finality-aware on-chain event processing, live LTV computation in basis points, DACS score updated on every price movement, and institution-authorized liquidation control — all automated. The always-on layer that makes crypto collateral operationally viable for credit union loan officers.

PATENT PENDING

NCUA Examiner View

Read-only audit dashboard scoped per credit union. Live on-chain collateral positions, LTV ratios, liquidation events, and full BSA/AML audit trail. Built for NCUA examination from day one.

Programmable Stablecoin Wallets

Embedded wallet infrastructure for programmable stablecoin disbursement wired end-to-end into the loan origination and settlement pipeline. USDC collateral flows, cross-chain transfer protocol, and programmatic wallet provisioning live in production — enabling stablecoin-denominated disbursement and cross-chain settlement without exposing credit unions to custodial complexity.

LIVE IN PRODUCTION

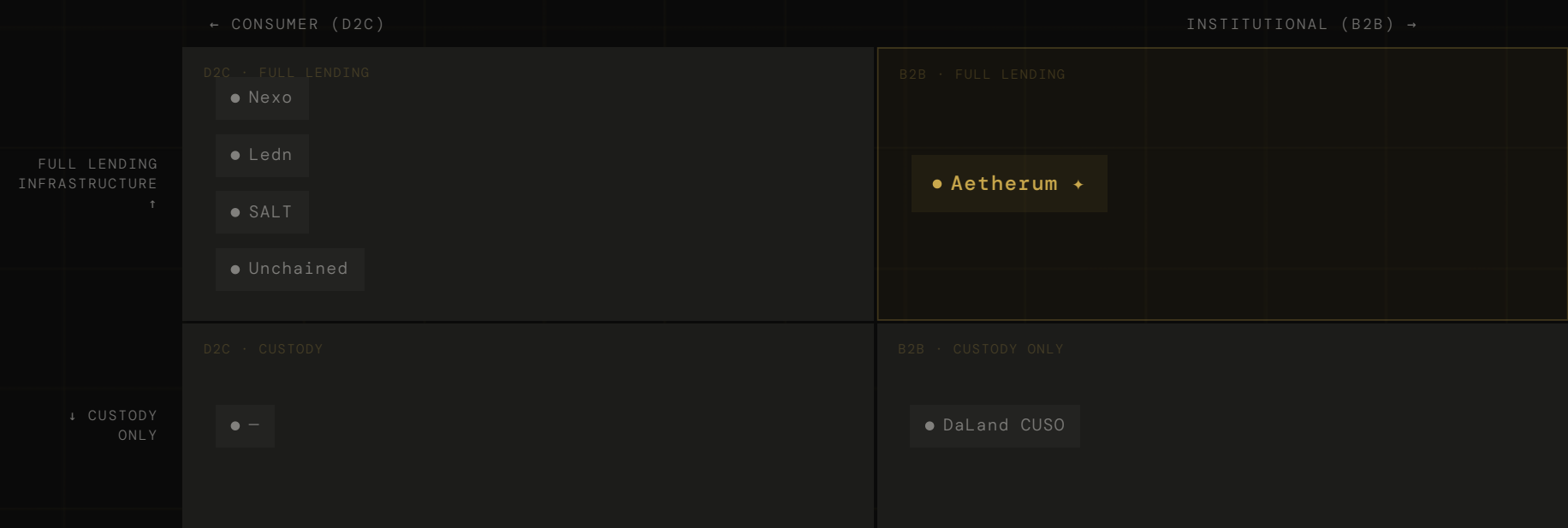
The *largest untapped* lending opportunity in credit union history.

"Credit unions are sitting on the largest untapped lending opportunity in their history. Aetherum is the infrastructure layer that makes it accessible."

TAM · TOTAL ADDRESSABLE MARKET \$150B+ Untapped Crypto Lending Gap 56M Americans hold crypto. Less than 1% can borrow against it through a regulated institution. That's a \$150B+ untapped lending gap — with no compliant path to liquidity today.	SAM · SERVICEABLE ADDRESSABLE MARKET \$15–25B NCUA-Regulated Credit Unions 4,800 NCUA-regulated credit unions. 140M members. An estimated 21M hold crypto today — with no compliant path to liquidity through their trusted financial institution.	SOM · SERVICEABLE OBTAINABLE MARKET \$1.2M → \$36M ARR 3-Year Ramp Year 1: 10 CUs × 100 loans/mo = \$1.2M ARR Year 2: 50 CUs × 150 loans/mo = \$9M ARR Year 3: 150 CUs × 200 loans/mo = \$36M ARR
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Built for institutions.

No one else is.



"Every competitor is building faster pipes. Aetherum is building the first crypto lending platform purpose-built for regulated institutions — with the compliance infrastructure to match."

Most fintech shows you a dashboard.
Aetherum shows you the reasoning.



DACS™ RISK GAUGE

Member sees their risk profile in real time
— not just a number.



SMART REVIEW™

Every loan comes with an explanation the
member can actually read.

LOAN OFFICER DASHBOARD

12 PENDING	\$1.4M VOLUME MTD	96% COMPLIANCE		
MEMBER	AMOUNT	DACS	KYC	STATUS
A. Reyes	\$24,500	81.8	✓ PASS	APPROVED
M. Chen	\$50,000	77.2	✓ PASS	REVIEW
T. Williams	\$12,000	62.4	✓ PASS	PENDING
S. Patel	\$95,000	88.1	✓ PASS	APPROVED
D. Kim	\$8,500	41.9	REVIEW	FLAGGED

LOAN OFFICER PORTAL

The CU sees everything their
examiner wants to see.

Built to the *compliance bar* that keeps others out.

BANK CONNECTIVITY	Financial data connectivity (CRA bundle) approved and live	— 6 core products active. Production approval used as investor data room leverage.	
COMPLIANCE LAYER	KYC/AML & fraud integration	— Executed MSA with production-ready compliance infrastructure. Annual billing. Live.	
3 CORE SYSTEMS	Jack Henry FIN contract executed. Banno plugin live in Garden sandbox. Corelation KeyStone integration built and demo-ready. Fiserv AppMarket listing in progress.		
LIQUIDATION LAYER	Three-tier liquidation escalation — human oversight for every large position	— Tier 1 auto-executes, Tier 2 pages risk officer with 30-min override window, Tier 3 requires explicit loan officer authorization. Twilio SMS + SendGrid alerts, loan officer dashboard actions, full audit trail. Built May 26, 2026.	LIVE
STABLECOIN SETTLEMENT	Stablecoin Settlement Infrastructure — live in production	— Three-layer settlement stack operational: USDC Collateral Flows, Cross-Chain Transfer Protocol, and Programmable Wallet Infrastructure — live in production.	LIVE
ARC TESTNET	CollateralEscrowTimelock deployed to Arc Testnet (Circle Layer-1, Chain ID: 5042002) <code>0x1fF9a4c2...8897</code> — USDC native gas token. Mainnet Q3 2026.		
CUSTODY	Enterprise custody — live	— Each CU holds its own direct account — Aetherum never touches assets. Non-custodial by architecture.	
MEMBER CRYPTO INSIGHTS	New intelligence layer	— Real-time visibility into member-owned digital assets — consent-based, examiner-ready, and patent pending.	

DACS™		Ten-pillar risk model, live gauge wired to API. Includes TPSOE as sixth pillar, CUMember as 10th, FHE-DACS as eighth, DACS-LTV as ninth, and XRPL-COLLATERAL™ as tenth provisional.	across DACS, TPSOE, FHE-DACS, CUMember, DACS-LTV, XLS-66, Key patents Governance, Collateral Liquidation Engine pending(AETHERUM-009), and Native LTV Monitoring (AETHERUM-010).	PATENT PENDING
TPSOE™		Tax-Position Score & Collateral Optimizer ranks pledge assets by embedded tax liability. No B2B competitor offers this to institutions.		PATENT PENDING
FHE-DACS		Homomorphic encryption layer for DACS risk assessment. Member financial data remains encrypted throughout the entire underwriting decisioning process.		PATENT PENDING
DACS-LTV™		Automated on-chain collateral LTV attestation triggered by DACS scoring. Collateral value and loan balance set directly on the AetherumCollateralLTVModule in real time.		PATENT PENDING
XRPL-COLLATERAL™		XLS-66 LoanBroker integration architecture for NCUA-regulated credit unions. Covers: DACS-gated LoanBroker provisioning, Broker-Initiated Dual-Signature Flow (BIDSF) for compliant LoanSet transaction authorization, First-Loss Capital management as a regulatory capital analog, GENIUS Act Freeze/Deep Freeze compliance controls mapping, and XLS-70 member credential integration with CIP recordkeeping timestamps. Live demo running on XRPL testnet.		PATENT PENDING
ECOSYSTEM ADVISOR		Steve Salzer, SVP at Velera — Board Advisor to Aetherum.	Chairs the Board of Ongoing Operations LLC and sits on the NACUSO Regulatory Advocacy Committee and CUNA Payments Subcommittee.	
SMART REVIEW		Shipped — live in production	— AI-powered underwriting guidance live in production, zero additional friction in the loan flow. Members receive a plain-language borrowing assessment automatically at the point of decision.	
RIPPLE / XRPL		XLS-66 LoanBroker	— Live integration running on XRPL testnet — LoanBrokerSet provisioning, dual-signature LoanSet flow (BIDSF), and First-Loss Capital health monitor all operational. Ripple BD relationship active — XLS-66 Design Partner designation in negotiation. RLUSD as native loan currency.	LIVE ON TESTNET
CIE™		Collateral Intelligence Engine — shipped today	— Real-time collateral surveillance pipeline live in production. Custody webhook → LTV computation → DACS score update → risk state transition → audit log. 5-minute price monitor scheduler watching all active positions. 3 database schemas, 6 REST endpoints, live Loan Officer Dashboard panel.	PATENT PENDING

NATIVE LTV MONITOR		Native real-time — BTC and ETH collateral wallets polled every 60 seconds via Blockstream and LTV monitoring — Alchemy. 30-day rolling volatility feeds DACS scoring. No third-party monitoring live in production dependency. AETHERUM-010 · Application #64/075,187 · Filed May 26, 2026.	LIVE
DDQ ENGINE		35-entry security questionnaire knowledge base — full SIG framework — Covers custody, encryption, incident response, key management, BSA/AML, nth-party risk, and all SIG security domains. Instant, compliance-vetted answers for credit union IT and risk teams — no sales call required.	LIVE
PLATFORM		MVP Live — Full-stack platform live at test-app.aetherum.ai — NestJS backend, React frontend, Python microservices, cloud database and cache across 6 cloud services. ERC-3643 compliance modules deployed on Sepolia.	
ERC-3643		Clean-room implementation of three on-chain compliance modules — deployed on Sepolia testnet. Patent-pending CUMember identity attestation built on the T-REX standard. Live on-chain compliance attestation via ERC-3643 — deployed to Sepolia testnet.	PATENT PENDING
JACK HENRY FIN		Contract executed. Banno plugin live in Jack Henry Garden sandbox. Production provisioning in progress. Core banking distribution layer serving 900+ credit unions.	
CORELATION KEYBRIDGE		Certified integration — live — Corelation KeyBridge KeyStone core banking API integration live. ~290 credit unions on KeyStone reachable from day one.	
RLUSD · RIPPLE		Supported collateral asset — XRPL integration live — RLUSD added as a supported collateral asset. Expands the addressable borrower base to XRP Ledger holders.	
PHAROS		Pharos Network · Atlantic Testnet · USDC + CCTP Live — Live integration with Pharos Atlantic Testnet. USDC contract wired, wallet generation, balance checker, and USDC disbursement demo powered by Circle CCTP.	
PHAROS · SPEED		Pharos Atlantic Testnet — USDC integration — Demonstrates Aetherum's ability to integrate emerging settlement infrastructure as it launches. Zero lag from announcement to live demo.	

ARC · CIRCLE	AetherumCollateralEscrow (ACE) — Arc Testnet — COMPLETE	— ERC-20 collateral escrow token deployed to Circle's Arc Testnet (May 23, 2026). Contract: 0xf7dff33870fafb6a453c1a43b0bd4d89d001a9ed. Demonstrates live readiness on Circle's Layer-1 settlement infrastructure.	↗ ArcScan
T-REX / ERC-3643	Only on-chain compliance layer purpose-built for credit union lending.		

07 · REVENUE MODEL

Five streams. *Platform-fee led.*

Revenue is tied directly to loan volume — meaning we grow as our credit union partners grow. The SaaS license layer adds predictable recurring revenue at scale.

ARR TARGET	$\$1\text{K/loan} \times 100 \text{ loans/mo} \times 10 \text{ CUs} = \1.2M ARR Tax Intelligence add-on layer creates additional recurring revenue per CU partner. ERC-3643 module licensing creates a second revenue surface independent of loan volume.
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70% Platform Fees	20% Transaction Fees
\$500–\$1,500 per loan originated. Primary revenue driver tied to volume.	0.5–1% on crypto collateral transactions processed through the platform.
10% SaaS License	New Tax Intelligence Module
\$50K–\$150K/year for large credit unions. Predictable ARR layer at scale.	\$10K–25K/year per credit union as a premium add-on. Tells members not just what they can borrow — but which assets to pledge and how much they save in taxes by borrowing instead of selling. No B2B

competitor offers this to institutions.

New

ERC-3643 Compliance Layer

Per-attestation module licensing fees. Any RWA issuer requiring US jurisdiction, collateral LTV gating, or CU member identity verification licenses Aetherum's compliance modules.

08 · TEAM & ADVISORS

Built by operators.
Advised by the ecosystem.

John Bailey

FOUNDER & CEO

20+ years in technical recruiting with placements at Google, Spotify, and Goldman Sachs. Previously co-founded Thorium Data (Data Catalog). Building Aetherum full-time since early

Steve Salzer

BOARD ADVISOR · SVP, VELERA

SVP at Velera since 2017. Chairs the Board of Managers at Ongoing Operations LLC. Sits on the NACUSO Regulatory Advocacy Committee and CUNA Payments Subcommittee. JD, University of Chicago Law School.

2025, managing all product, BD, fundraising, and technical implementation.							
Rahul Goel ADVISOR · JP MORGAN / UBER Cybersecurity and financial infrastructure expertise from senior roles at JP Morgan and Uber, including involvement with Meta's Novi digital wallet. Advises on platform security architecture and institutional trust.				Saiprasad Raut ADVISOR · WORLDPAY / VANAR Head of Payments Infrastructure at Vanar, formerly Worldpay. Deep payments and blockchain infrastructure expertise guiding Aetherum's settlement and custody architecture.			

09 · THE ASK

Closing a *\$1M Pre-Seed* Round.

We're in the final stretch of closing our pre-seed round. Proceeds go toward closing the first two credit union pilots, completing platform development, and bringing on a CTO.

Capital deploys us to first 3 credit union pilots — reducing market risk and proving the explanation layer at scale.

ROUND SIZE	\$1,000,000
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REQUEST A DEMO

Let's talk.

20 minutes. We'll walk through the platform live and answer any diligence questions.

BOOK A CALL

